VEDANT DUBEY

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Electronics and Communication Engineering, National Institute of Technology, Raipur

EDUCATION

National Institute of Technology, Raipur *2023 – 2027 (Expected)*
B.Tech in Electronics and Communication Engineering

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, SQL, TypeScript
Frameworks: React.js, Next.js, Node.js, Flask, PyQt, Express.js
AI/ML: LangChain, HuggingFace Transformers, PyTorch, Pinecone, RAG, LLMs, Fuzzy Logic
Backend: REST APIs, GraphQL, Prisma, Supabase, PostgreSQL,MongoDB, Authentication (JWT/OAuth)
Developer Tools: Git, GitHub, Linux, Playwright, OpenFOAM, PyVNT, Jest, Pytest, CI/CD Pipelines, Docker, AWS (EC2, S3)

WORK EXPERIENCE

Cubelelo — Software Engineer Intern (Currently Part-Time) *May 2026 – Present*
Raipur, India

- Built a rules-based classification engine to auto-categorise returns across 10+ marketplace integrations, replacing manual triage.
- Designed a normalised\ Postgres schema (Supabase) to reconcile order and inventory state across 12+ marketplace APIs with differing rate limits and pagination, using idempotent upsert jobs to prevent duplicate records on retry.
- Built Playwright-based automation for marketplaces lacking public APIs, handling session persistence and scheduled polling to replace manual spreadsheet reconciliation, saving ~25 hours/week of manual work.
- Deployed 15+ backend services in Next.js and Supabase for real-time warehouse ops, barcode scanning, and inventory tracking.
- **Tech Stack:** Next.js, TypeScript, SQL, Supabase, REST APIs, GraphQL, Playwright

FOSSEE, IIT Bombay — Part-Time Software Developer, CFD (OpenFOAM) & LLM Integration | [Code] *May 2025 – Present*
Remote

- Built a parser converting 100+ OpenFOAM case configurations into structured PyVNT trees, using a grammar-based extraction layer to handle inconsistent formatting across case files.
- Implemented bi-directional translation across 5+ OpenFOAM config formats, achieving 100% data fidelity.
- Engineered an RAG pipeline to interpret free-form CFD config comments and map them to structured PyVNT fields, improving field-extraction accuracy by 35% over regex-only baseline.
- Automated CFD workflow generation, reducing setup time by 50% and eliminating configuration errors.
- **Tech Stack:** Python, OpenFOAM, PyQt, PyVNT, HuggingFace Transformers, LLMs

PROJECTS

FlowOps — Engineering Intelligence SaaS Platform | [Live] | [Code] *November 2025 – Present*

- Built a multi-tenant SaaS platform (Next.js, Express, PostgreSQL/Prisma) that ingests GitHub commits, pull requests, and reviews via OAuth and webhooks to surface 10+ real-time engineering metrics.
- Automated AI review pipeline (via Gemini/LangChain), including security and anti-pattern detection, generating refactors
- Designed a PR automation engine that detects stale and unreviewed PRs, auto-approves low-risk changes, and delivers nudges via email and Slack, with a weekly "time saved" impact digest for engineering managers, reducing PR turnaround time by 40%.
- Established secure GitHub OAuth with httpOnly-cookie session authentication and Socket.io for real-time dashboard updates across a multi-tenant organization model with 4-tier RBAC.
- **Tech Stack:** Next.js, Node.js, Express, PostgreSQL, Prisma, Docker, GitHub Actions, GitHub OAuth, REST APIs

CookGPT — Multimodal RAG LLM Chatbot (1st Place) | [Live] | [Code] *March 2025*

- Developed a multimodal RAG chatbot capable of processing text and image queries with sub-second response latency.
- Built a Flask backend using LangChain and Pinecone to perform semantic retrieval over 10,000+ embeddings.
- Integrated the Gemini API to enhance contextual reasoning, multimodal understanding, and response accuracy, boosting query response accuracy by 30%.
- Crafted a scalable, responsive frontend using React with Clerk-based authentication and role management, and session persistence.
- **Tech Stack:** Python (Flask, LangChain, Pinecone, Gemini API, Scikit-learn), React.js, TailwindCSS, Clerk

PUBLICATIONS AND RESEARCH

Multi-Actor DRL with LLM for Sparse Hybrid RIS-IAB Networks | [Paper] *Submitted, Under Review*

- Co-authored an IEEE research paper proposing a Multi-Actor Deep Reinforcement Learning framework combining DistilBERT and DDPG for Sparse Hybrid RIS-IAB networks, improving energy efficiency by 17.4% and 42.1%.

ACHIEVEMENTS

- **Winner**, Hack-o-Harbour 2025 — IIIT Naya Raipur (150+ teams), American Allegiance of Education Hackathon (150+ teams)
- **Runner-Up**, Codeutsav 8.0 — Central India's largest hackathon, hosted by NIT Raipur (300+ teams)
- 450+ DSA problems solved (LeetCode, CodeChef); Selected for **Amazon ML Summer School** 2026